

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
- Summary of all results

Introduction

- The commercial space industry includes companies like Virgin Galactic, Rocket Lab, Blue Origin, and SpaceX, with SpaceX being notable for its achievements and cost-effective rocket launches.
- SpaceX's Falcon 9 rocket launches cost about \$62 million, significantly less than competitors, largely due to the ability to reuse the first stage of the rocket.
- Instead of traditional rocket science, machine learning models will be trained on public data to predict whether the first stage will land successfully, aiding in cost estimation and competitive bidding.

Section 1

Methodology

Methodology

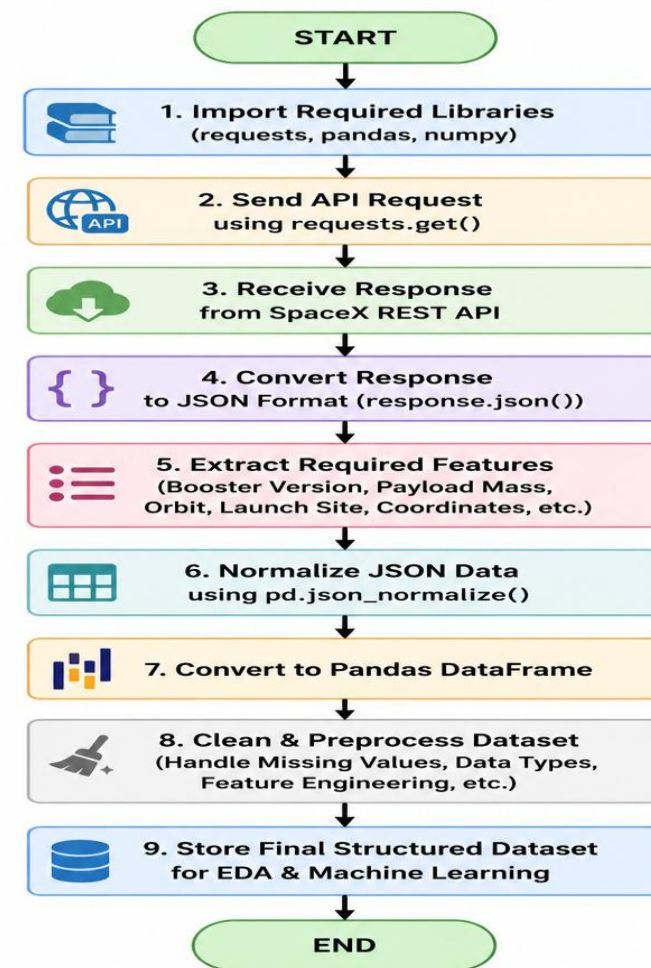
Executive Summary

- Data collection methodology:
 - Describe how data was collected
Data was first collected by importing the requests library. By using the .get method, data related to various columns like booster version, lat, long, name, payload mass, were extracted, converted to json. Later normalized to a dataframe by json_normalize().
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection – SpaceX API

- **Data Collection using SpaceX REST API**
- Data was collected using the SpaceX REST API through Python's requests library.
- The .get() method was used to retrieve launch-related information such as:
 - Booster Version
 - Payload Mass
 - Launch Site
 - Latitude & Longitude
 - Rocket Name
 - Orbit Type
 - Flight Number
- The retrieved API response was initially in JSON format.
- The JSON data was then normalized and converted into a structured Pandas DataFrame using:
 - `pd.json_normalize()`
- Further preprocessing and feature extraction were performed to prepare the dataset for exploratory data analysis and machine learning tasks.

[GitHub](#)

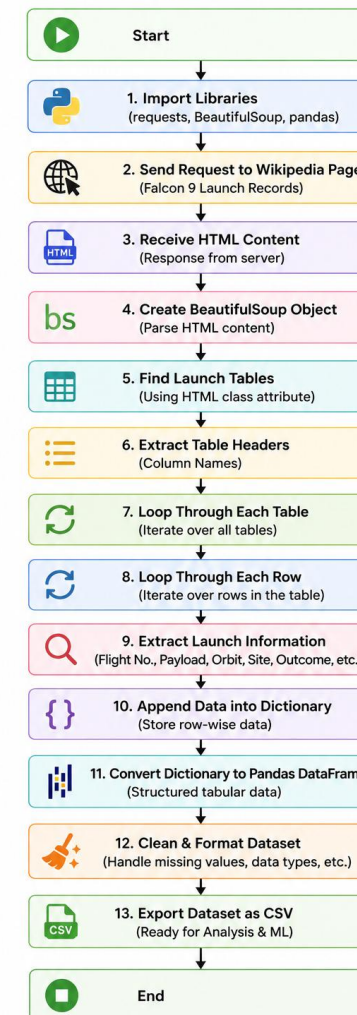


Data Collection - Scraping

After creating the **BeautifulSoup** object, all HTML tables present on the webpage were identified. A **loop** was used to iterate through the tables and extract the required table headers (column names). Another loop was implemented to extract row-wise data from the table. The extracted information was appended into a dictionary and later converted into a structured **Pandas DataFrame** for further analysis and preprocessing.

[GitHub](#)

Web Scraping Flowchart

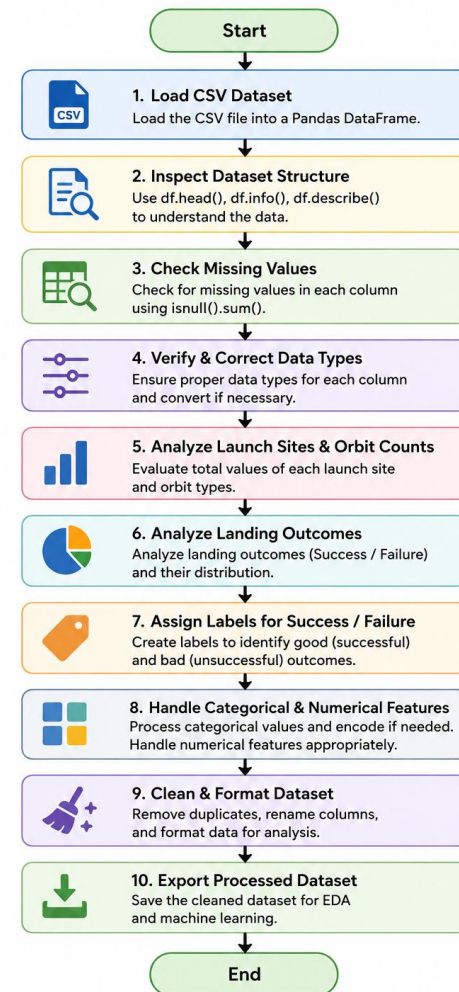


Data Wrangling

The collected CSV dataset was loaded into a **Pandas DataFrame** for preprocessing and analysis. **Missing values and data types** of each column were examined and **handled** appropriately. **Launch site statistics, orbit frequencies, and landing outcome distributions** were analyzed to better understand the dataset. **Labels** were assigned to classify successful and unsuccessful landing outcomes. Additional **preprocessing steps** such as **feature selection, formatting, and categorical value** handling were performed to prepare the dataset for **exploratory data analysis and machine learning models**. The **cleaned dataset** was then exported for further visualization and predictive modeling tasks.

[GitHub](#)

Data Wrangling Process Flowchart



EDA with Data Visualization

Exploratory Data Analysis was performed to identify **patterns, relationships, and trends** within the SpaceX launch dataset. **Scatter plots** were used to analyze relationships between variables such as :

Payload Mass vs Flight Number ,
Payload Mass vs Launch Site,
Flight Number vs Orbit Type
Payload Mass vs Orbit Type.

These scatter plots helped visualize how payload capacity, orbit selection, and launch frequency **influence mission outcomes**. **Bar charts** were created to compare the **success rates across different orbit types** and **launch sites, making categorical comparisons easier** to interpret. **Line charts** were used to visualize launch **success trends over time** and identify **improvements** in mission success rates across different years. The visualizations provided **meaningful insights** into launch **performance, payload behavior, and operational trends**.

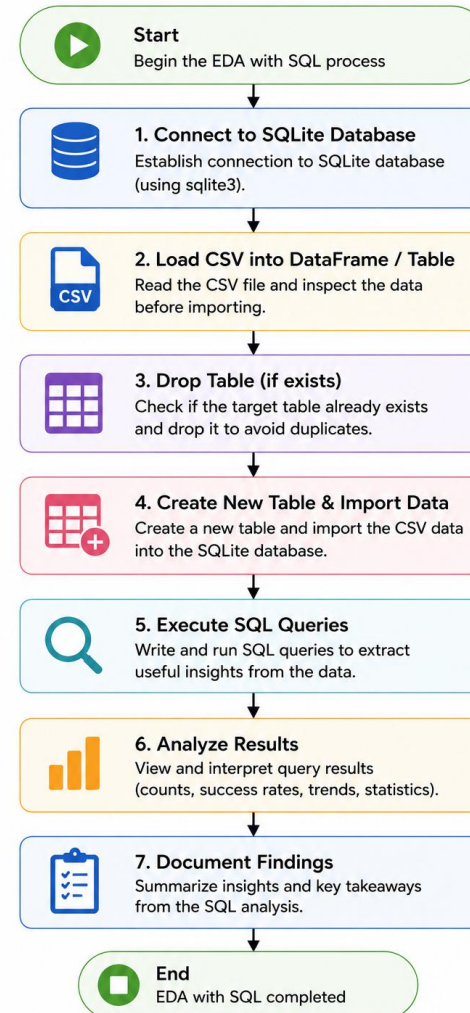
EDA with Data Visualization – Flowchart



EDA with SQL

- SQL was used to perform Exploratory Data Analysis (EDA) on the SpaceX launch dataset stored in a SQLite database. The CSV dataset was first loaded into a Pandas DataFrame and then imported into a SQLite database for querying and analysis. A database connection was established using `sqlite3`, enabling SQL operations on the dataset. Existing tables were dropped (if present) to avoid duplicate entries and ensure clean data insertion before creating new tables. Multiple SQL queries were executed to extract meaningful insights from the dataset, including: Total number of launch records, Distinct launch sites, and their frequencies, Orbit-wise launch success rates, Launch site performance comparison, Year-wise launch trends, Payload mass statistics and aggregations, SQL aggregation functions such as `COUNT()`, `AVG()`, `GROUP BY`, `ORDER BY`, and filtering conditions using `WHERE` were applied to analyze patterns and trends. The query results helped identify operational trends, success distributions, and relationships between launch parameters. The SQL-based analysis provided structured insights that supported further visualization and machine learning stages of the project. [GitHub](#)

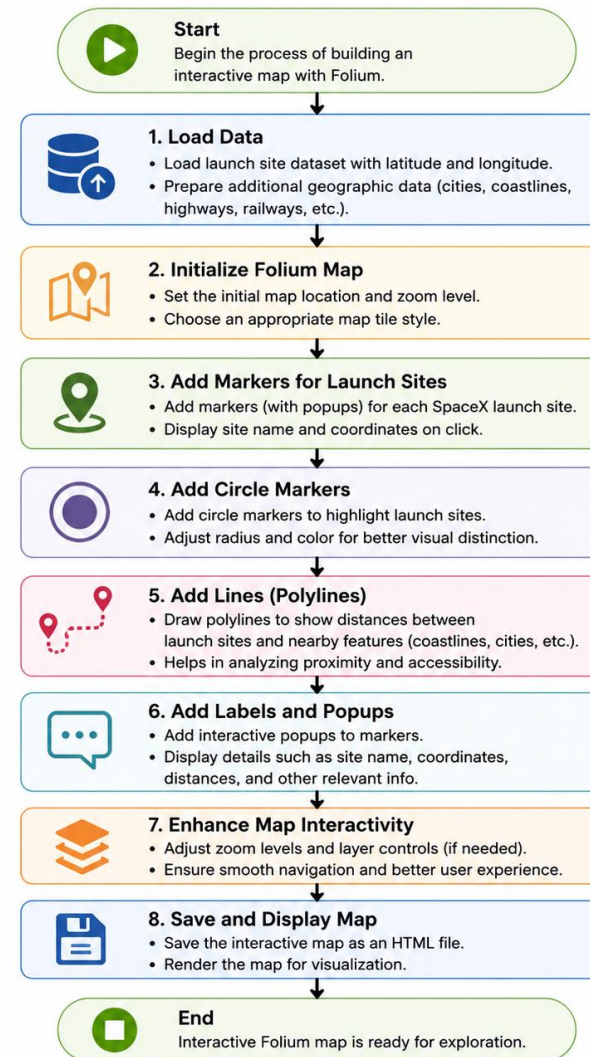
EDA with SQL – Process Flow



Build an Interactive Map with Folium

- An interactive map was created using the Folium library to visualize SpaceX launch site locations and nearby geographical features. Map objects such as markers, circles, labels, and polylines were added to improve geographical interpretation and interactivity. Markers were used to indicate the exact locations of SpaceX launch sites along with their corresponding coordinates and site names. Circle markers were added to visually highlight launch sites and distinguish them from surrounding map elements. Distance lines (polylines) were drawn between launch sites and nearby landmarks such as coastlines, highways, railways, and cities to analyze location advantages and accessibility. Interactive labels and popups were included to display additional information such as launch site names, coordinates, and calculated distances. Different map layers and zoom levels were used to improve navigation and provide better spatial understanding of launch operations. The interactive Folium visualization helped analyze how geographical positioning may influence launch operations, recovery logistics, and mission planning. [GitHub](#)

Build an Interactive Map with Folium – Process Flow

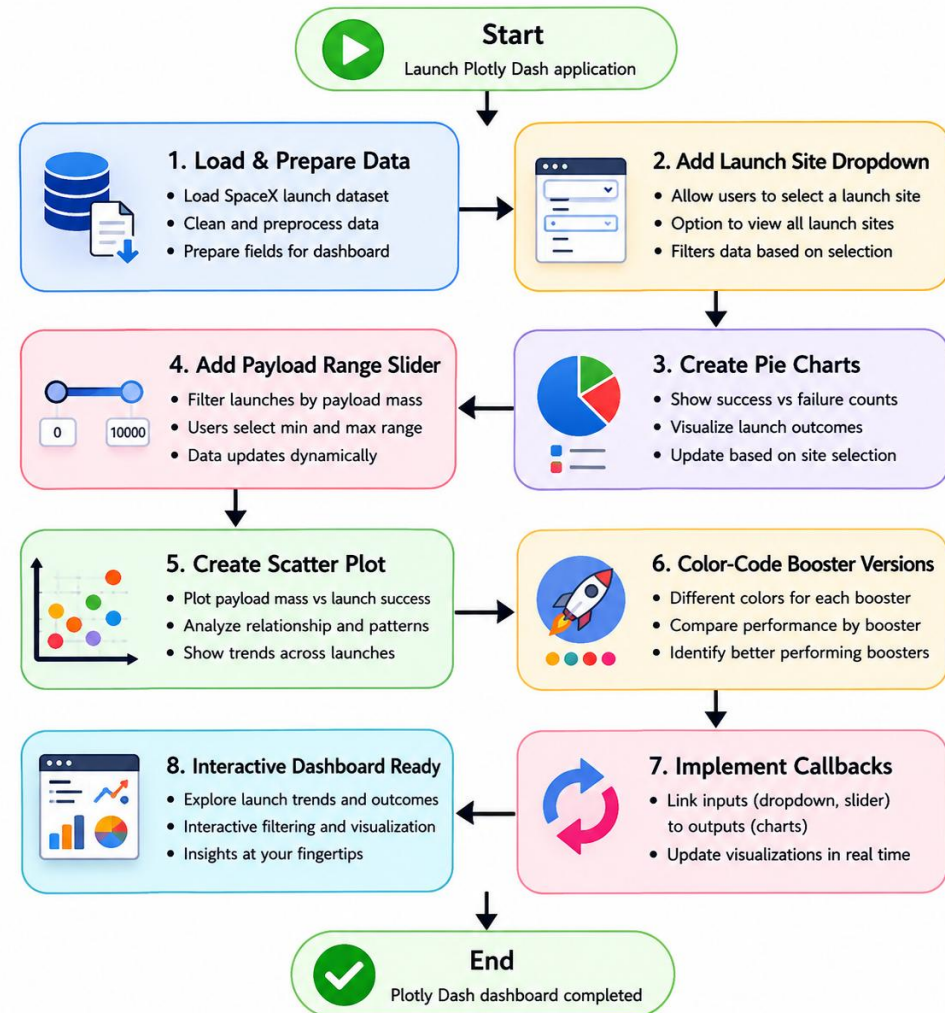


Build a Dashboard with Plotly Dash

- Developed an interactive dashboard using Plotly Dash to visualize SpaceX launch data.
- Added a launch site dropdown menu to filter and analyze launches by site.
- Created pie charts to display launch success and failure distributions.
- Implemented a payload range slider to dynamically filter payload mass values.
- Built scatter plots to visualize the relationship between payload mass and launch success.
- Used color-coded booster versions in scatter plots to compare rocket performance.
- Added interactive callbacks to update graphs automatically based on user selections.
- Enabled real-time exploration of launch trends, payload behavior, and mission outcomes.

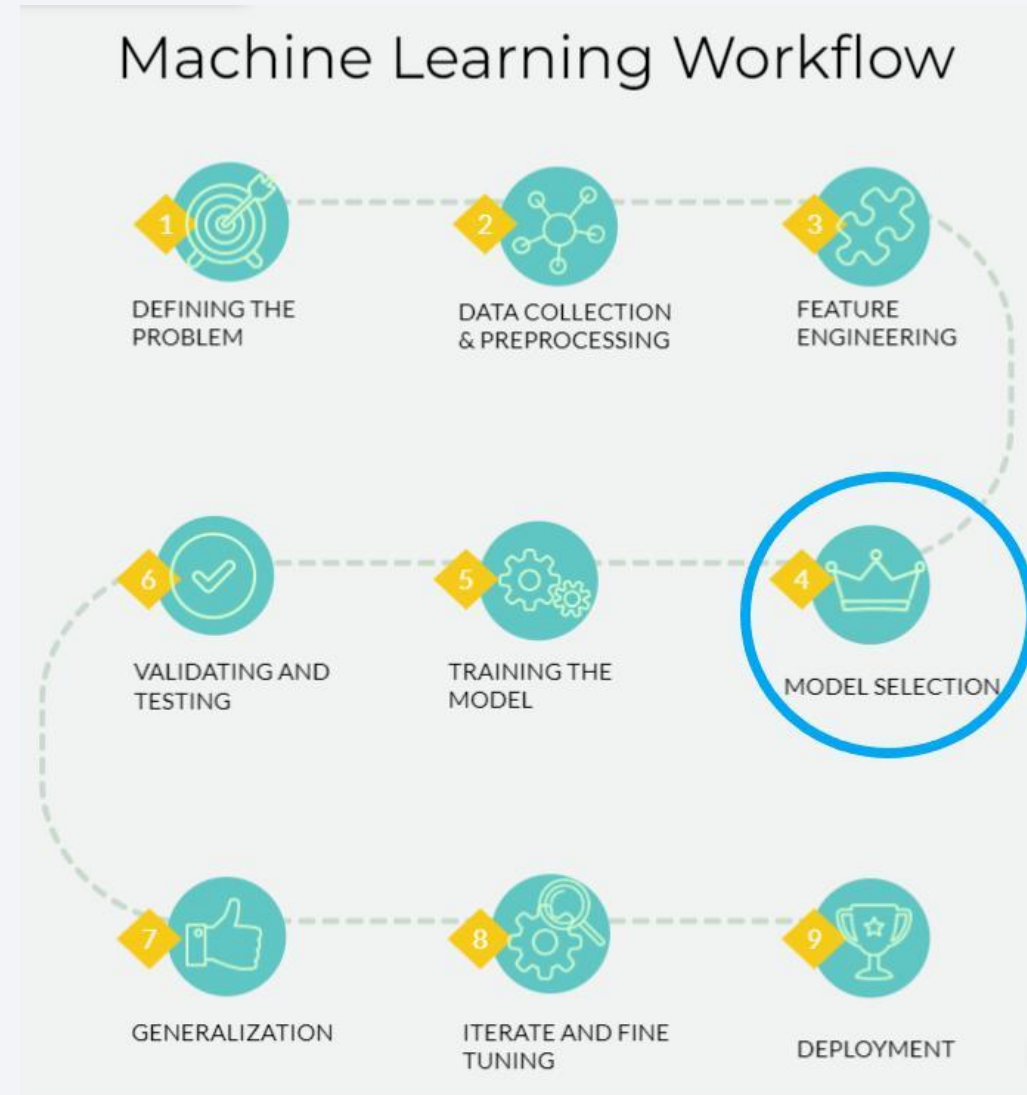
[GitHub](#)

Build a Dashboard with Plotly Dash – Flowchart



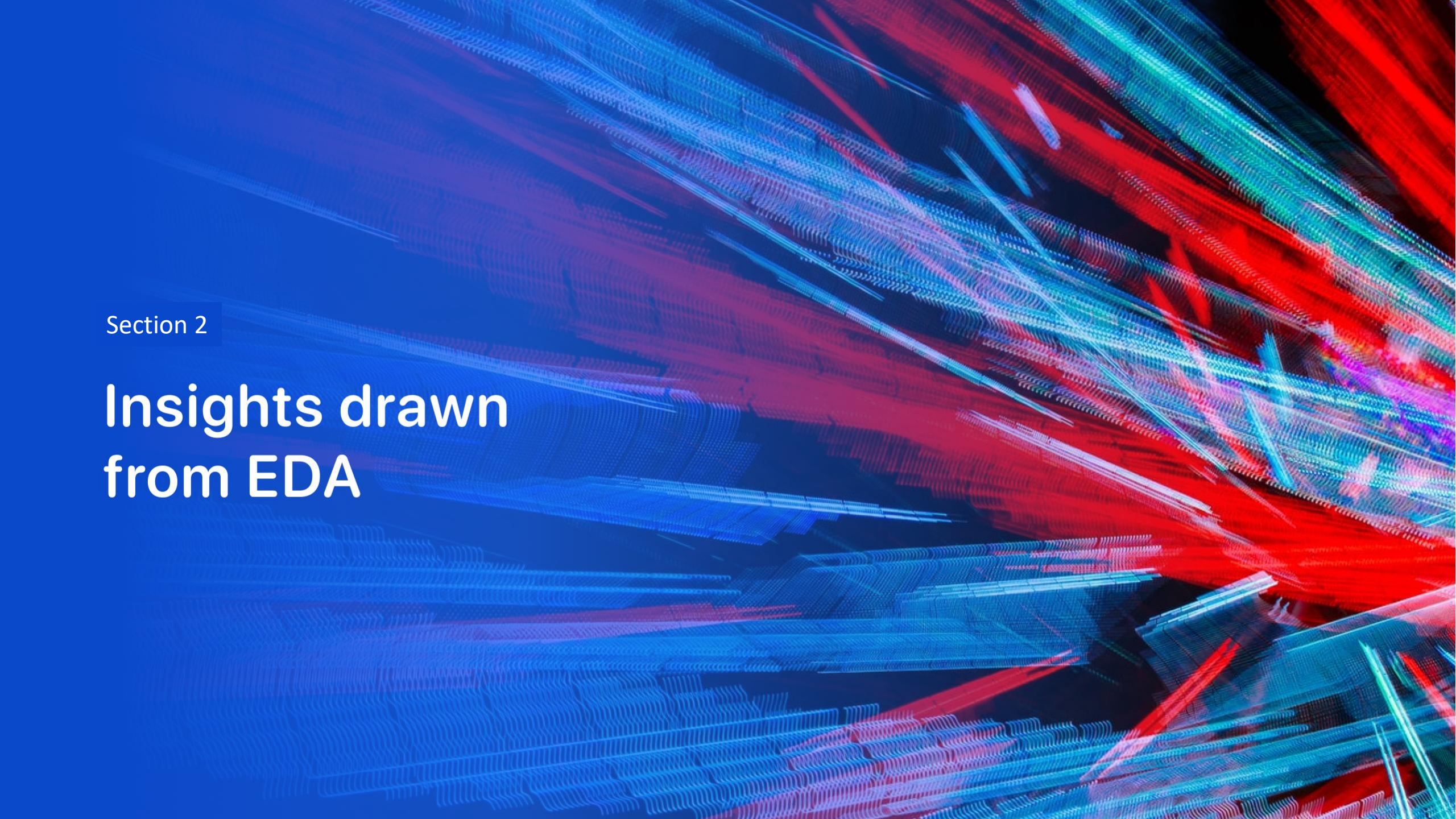
Predictive Analysis (Classification)

- Developed machine learning classification models to predict SpaceX launch success.
- Preprocessed the dataset using feature selection, one-hot encoding, and data standardization.
- Split the dataset into training and testing sets for model evaluation.
- Implemented multiple classification algorithms:
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Decision Tree
 - K-Nearest Neighbors (KNN)
- Performed hyperparameter tuning using
- Evaluated models using accuracy scores and confusion matrices.
- Compared model performances to identify the best-performing classifier.
- Analyzed prediction outcomes using confusion matrix metrics.
- Selected the optimal model based on validation and test accuracy.



Results

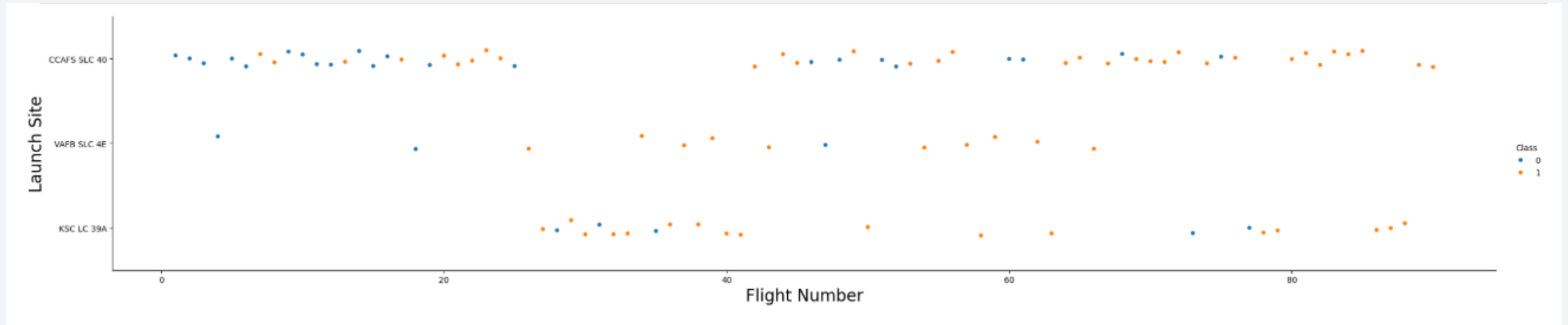
Exploratory Data Analysis (EDA) revealed important trends between payload mass, orbit type, launch sites, and mission outcomes. Visualization analysis showed that payload mass and orbit selection significantly influenced launch success rates. Interactive dashboards enabled dynamic filtering and real-time analysis of launch site performance and payload behavior. Folium-based geographical analysis highlighted the strategic positioning of launch sites near coastlines and transportation routes. Machine learning classification models successfully predicted launch outcomes using historical mission data. Multiple models were compared, including Logistic Regression, SVM, Decision Tree, and KNN classifiers. Hyperparameter tuning improved model performance and helped identify the most accurate predictive model. Confusion matrices and accuracy scores were used to evaluate classification effectiveness and prediction reliability. The project demonstrated how data science, visualization, and machine learning can support aerospace mission analysis and decision-making.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

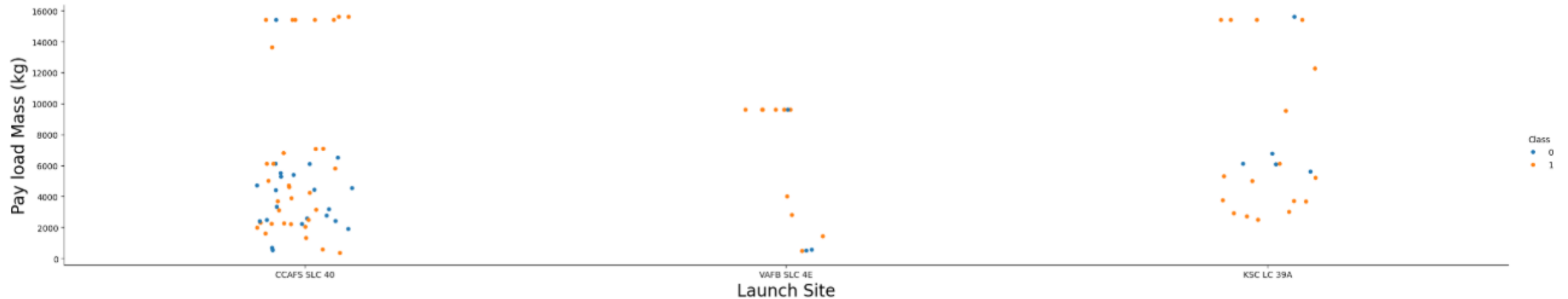
Insights drawn from EDA

Flight Number vs. Launch Site



- The scatter plot illustrates the relationship between Flight Number and Launch Site across different SpaceX missions.
- Distinct flight number ranges appear to be associated with specific launch sites:
Flight numbers between 0–20 were predominantly launched from CCAFS SLC 40. Flights in the 25–40 range were more frequently launched from KSC LC 39A. Flight numbers between 40–80 again show a strong concentration around CCAFS SLC 40. VAFB SLC 4E appears less frequently compared to the other launch sites, indicating fewer missions conducted from that location.
- The distribution suggests that launch site usage changed over different mission phases and operational periods.
- However, there is little to no direct correlation between Flight Number and Launch Site, since launch site selection depends more on mission requirements, orbit type, payload characteristics, and operational constraints rather than the sequence of flights alone.

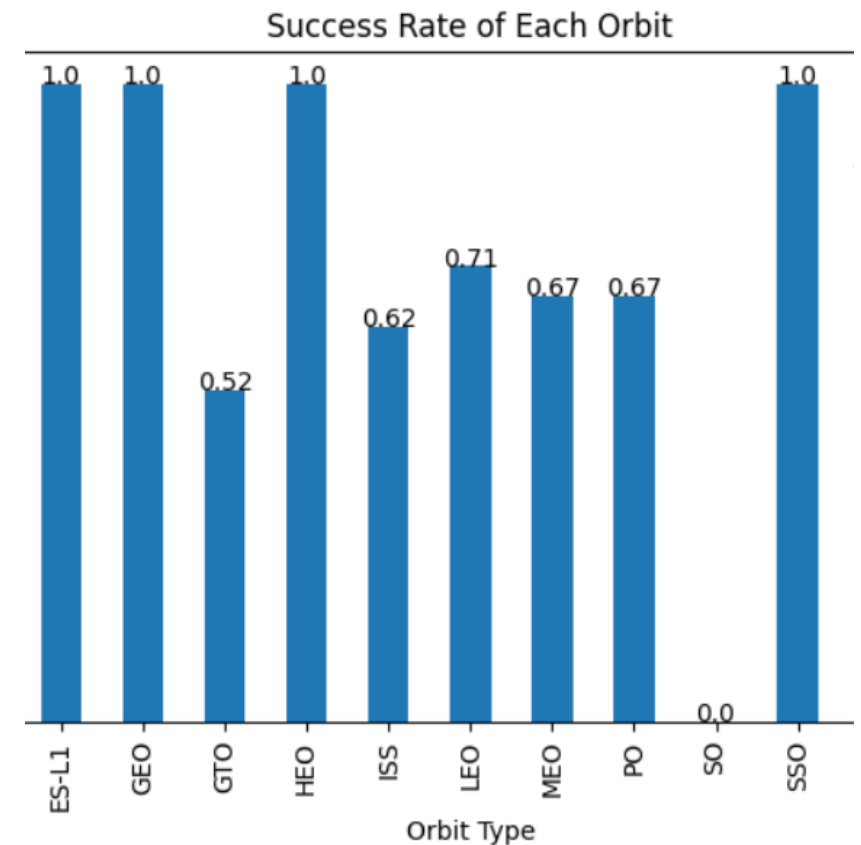
Payload vs. Launch Site



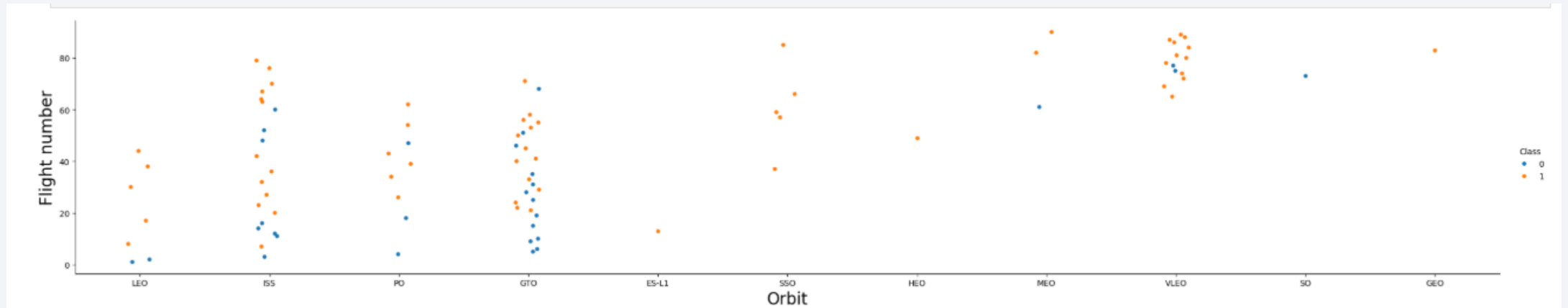
- The scatter plot illustrates the relationship between Payload Mass and Launch Site across different SpaceX missions.
- VAFB SLC 4E shows very limited launches involving extremely high payload masses, indicating lower suitability for heavy payload missions.
KSC LC 39A demonstrates better performance for high payload launches, with several missions carrying payloads exceeding 10,000 kg.
CCAFS SLC 40 contains a large concentration of low and moderate payload launches, suggesting it is frequently used for lighter missions.
- Based on the payload distribution:
CCAFS SLC 40 appears best suited for low payload missions. VAFB SLC 4E is more commonly associated with medium payload launches. KSC LC 39A is most effective for high payload missions. The variation in payload distribution across launch sites suggests that launch site selection depends heavily on mission requirements, rocket configuration, and payload capacity constraints.

Success Rate vs. Orbit Type

- The bar chart compares the launch success rates across different orbit types. Although ES-L1, GEO, and HEO display a 100% success rate, these results are based on only one successful launch attempt each, making the sample size too small for strong reliability conclusions.
- SSO also achieved a 100% success rate, but unlike the previous orbit types, it had multiple successful launch attempts, indicating stronger operational consistency.
- Surprisingly, GTO and ISS missions recorded comparatively lower success rates of approximately 52% and 62% respectively, despite being among the most frequently used orbit categories.
- The high number of launches for GTO and ISS suggests these orbit types are operationally important and commonly required, even with moderate success rates.
- VLEO achieved one of the strongest overall performances, maintaining an approximate 86% success rate across a relatively larger number of launches.
- Additional exploratory data analysis (EDA) was performed to calculate and compare the total number of launches associated with each orbit type. The analysis highlights that success percentage alone is not always sufficient for evaluation; the total number of launch attempts must also be considered for meaningful interpretation.

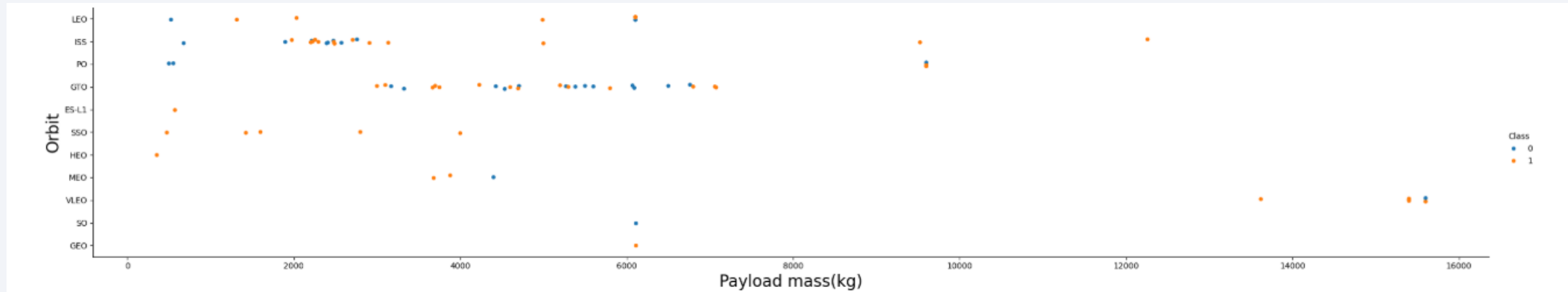


Flight Number vs. Orbit Type



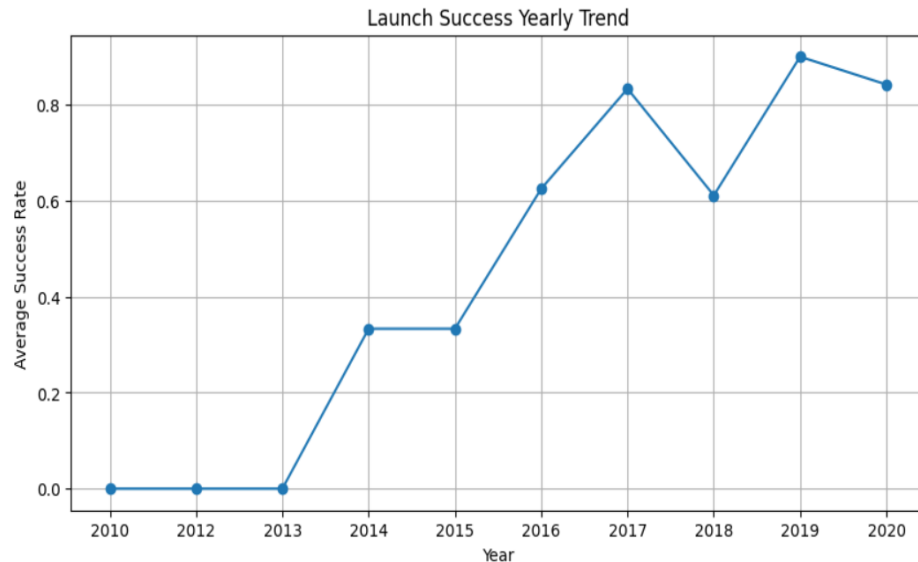
- The scatter plot illustrates the relationship between Flight Number and Orbit Type across different SpaceX missions. GTO and ISS orbit types appear across a wide range of flight numbers, indicating that these missions were consistently used throughout multiple launch phases and mission periods.
- VLEO missions are concentrated primarily in higher flight numbers, especially beyond flight number 70–80, suggesting that these missions were introduced more frequently in later operational stages. Orbit types such as ES-L1, HEO, SO, and GEO appear only a few times in the dataset, indicating limited mission usage or specialized operational requirements.
- The concentration of certain orbit types within specific flight number ranges suggests that SpaceX mission objectives evolved over time. Frequently occurring orbit types like GTO and ISS demonstrate operational flexibility and broader mission applicability compared to the less common orbit categories. The analysis also indicates that some orbit types are highly specialized and therefore naturally associated with fewer launches.

Payload vs. Orbit Type



- The scatter plot illustrates the relationship between Payload Mass and Orbit Type across different SpaceX missions. GTO missions demonstrate the ability to handle a wide range of payload masses, indicating high operational flexibility for medium to heavy payload launches. LEO missions are generally associated with lower payload masses, suggesting their suitability for lighter missions and short-range orbital deployments.
- VLEO missions are mostly concentrated in the higher payload mass range, indicating stronger compatibility with heavier payload operations. Several orbit types contain only a limited number of launch records, making it difficult to establish strong statistical conclusions. The distribution of payload masses across orbit categories appears scattered, suggesting little to no clear correlation between Payload Mass and Orbit Type. Additional operational factors such as mission objectives, rocket configuration, customer requirements, and orbital parameters likely influence orbit selection more significantly than payload mass alone. Therefore, the scatter plot alone is not sufficient to conclusively determine the impact of payload mass on orbit selection

Launch Success Yearly Trend



- The line chart illustrates the yearly trend of average SpaceX launch success rates from 2010 to 2020. The launch success rate remained relatively low during the early years (2010–2013), indicating the developmental and experimental phase of operations. A noticeable improvement began around 2014, where the success rate increased and temporarily stabilized, possibly representing a transitional phase in launch operations and technological refinement. From 2015 onwards, the overall success trend increased significantly, demonstrating improvements in rocket reliability, mission planning, and operational efficiency.
- The success rate experienced a temporary decline around 2018, indicating a short-term fluctuation or operational setback. However, the success rate recovered strongly in 2019, achieving one of the highest overall success levels in the dataset. The consistently high success rates in the later years suggest substantial advancements in SpaceX launch technology and mission execution capabilities. Overall, the trend indicates continuous long-term improvement in launch reliability despite occasional fluctuations.

All Launch Site Names

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

```
%sql select distinct Launch_Site from  
SPACEXTABLE;
```

Different launch sites were retrieved from the table using the SQL SELECT DISTINCT query, which returns only unique values from the Launch_Site column.

Launch Site Names Begin with 'CCA'

Records from launch sites beginning with CCA were retrieved using the SQL LIKE operator with the pattern 'CCA%', while LIMIT 5 was used to display only the first five matching rows.

```
%sql select * from SPACEXTABLE where Launch_Site LIKE 'CCA%' LIMIT 5;
```

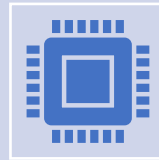
Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

SUM(PAYLOAD_MASS_KG_)
45596



```
%sql SELECT  
SUM(PAYLOAD_MASS__KG_) FROM  
SPACEXTABLE WHERE Customer =  
'NASA (CRS)';
```



The SQL query calculates the total payload mass delivered for missions where the customer was NASA (CRS) by using the aggregate function SUM() on the PAYLOAD_MASS__KG_ column.

Average Payload Mass by F9 v1.1

- %sql SELECT
AVG(PAYLOAD_MASS_KG_) FROM
SPACEXTABLE WHERE Booster_Version
LIKE 'F9 v1.1%';
- The query uses the aggregate function
AVG() to calculate the average payload
mass carried by the booster version F9
v1.1, filtering the records using the LIKE
operator.

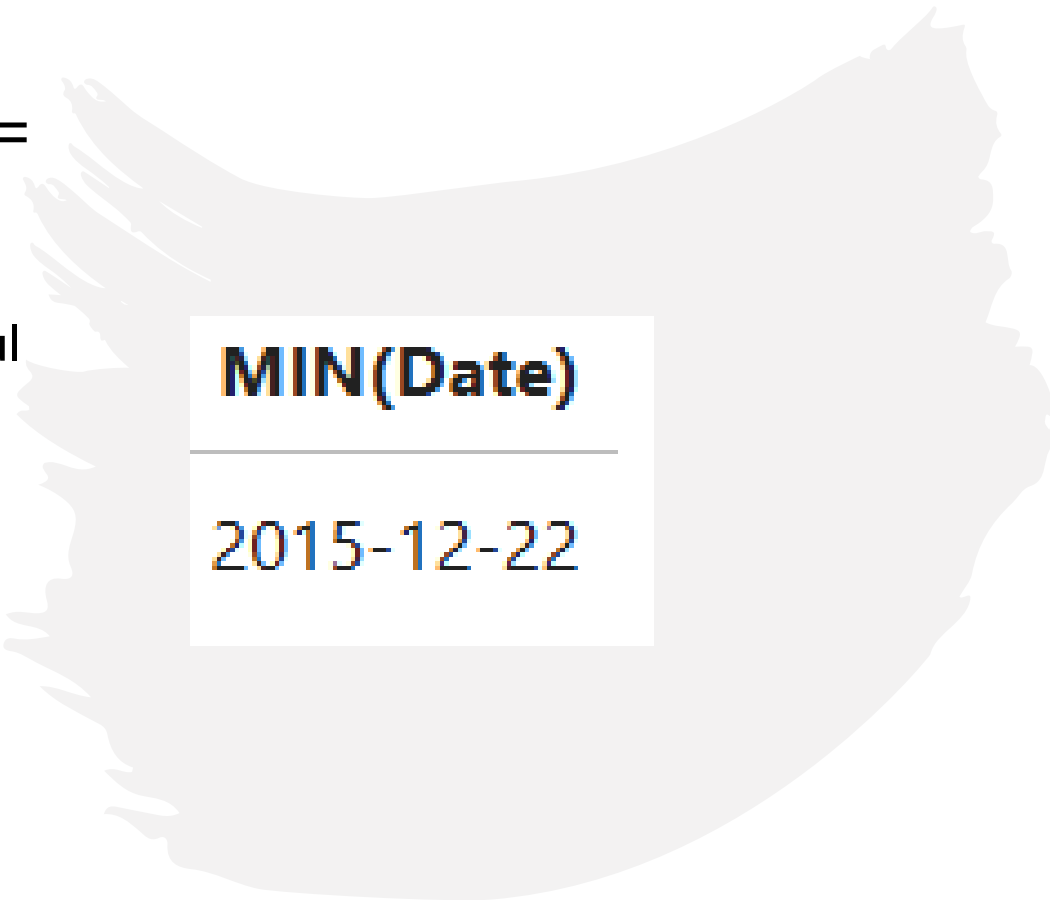


AVG(PAYLOAD_MASS_KG_)
2534.6666666666665

First Successful Ground Landing Date

```
%sql SELECT MIN(Date) FROM  
SPACEXTABLE WHERE Landing_Outcome =  
'Success (ground pad)';
```

The query retrieves the earliest date of a successful ground pad landing using the aggregate function MIN() on the Date column, filtered by the landing outcome Success (ground pad).



MIN(Date)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- `SELECT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;`
- Filters the dataset to find specific Falcon 9 booster versions that successfully landed on a drone ship while carrying a medium-heavy cargo weight between 4,000 and 6,000 kg.

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- `SELECT Mission_Outcome, COUNT(*) AS Total_Count FROM SPACEXTBL GROUP BY Mission_Outcome;`
- Groups the data by the overall mission completion status to count and compare how many total launches succeeded versus how many failed.

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload



```
SELECT BOOSTER_VERSION FROM  
  SPACEXTBL WHERE  
  PAYLOAD_MASS_KG_ = (SELECT  
    MAX(PAYLOAD_MASS_KG_) FROM  
    SPACEXTBL);
```



```
USES A NESTED SUBQUERY TO IDENTIFY  
  THE HIGHEST PAYLOAD WEIGHT  
  RECORDED IN THE TABLE (15,600 KG)  
  AND LISTS ALL THE BOOSTER VERSIONS  
  THAT CARRIED THAT MAXIMUM  
  CAPACITY.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- `SELECT Booster_Version, Launch_Site, Landing_Outcome FROM SPACEXTBL WHERE Landing_Outcome = 'Failure (drone ship)' AND Date LIKE '%2015%';`
- Extracts the exact booster names and sites for drone ship landing failures specifically during the year 2015 by using text pattern matching on the date column

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- `SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;`
- Aggregates and counts each type of landing result within a specific early historical timeframe, ranking them from most frequent to least frequent. (Note: Depending on how your SQLite strings are formatted, dates are sometimes queried using substr string functions if they were imported as DD/MM/YYYY).

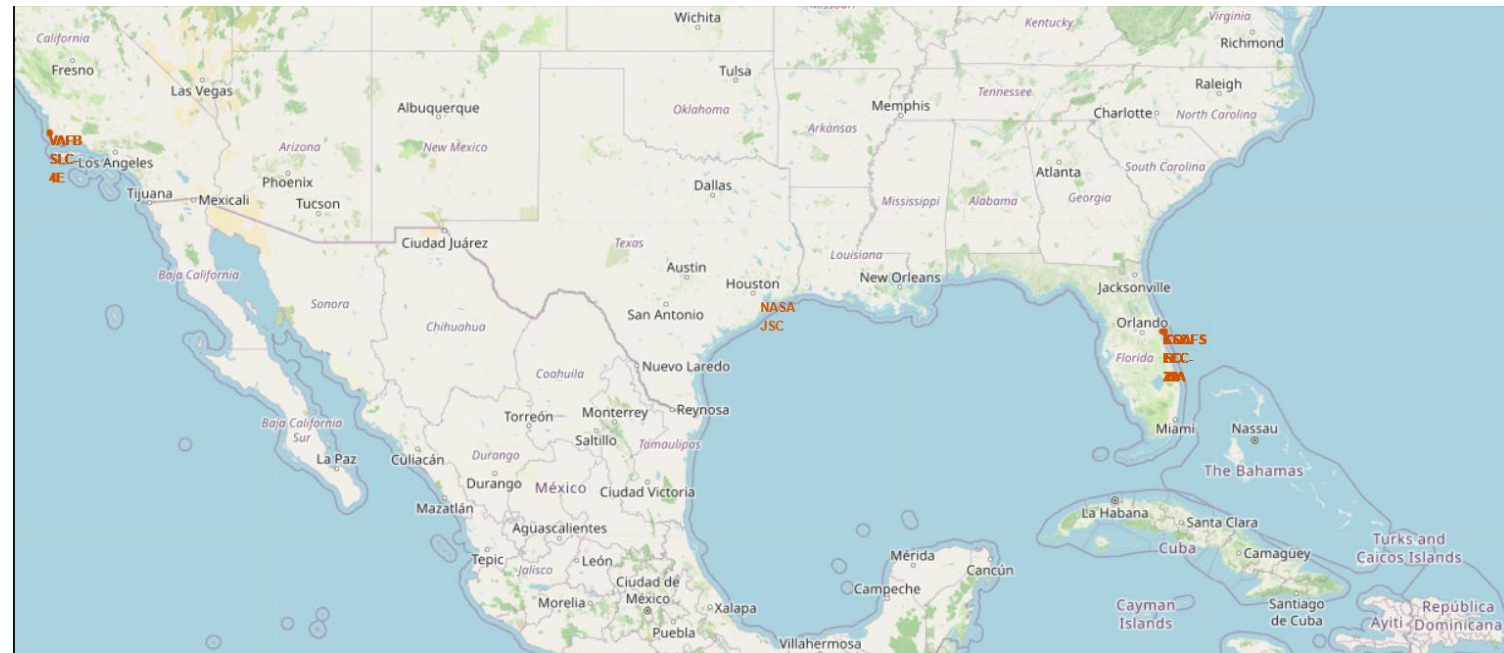
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left portion shows a clear, dark blue sky.

Section 3

Launch Sites Proximities Analysis

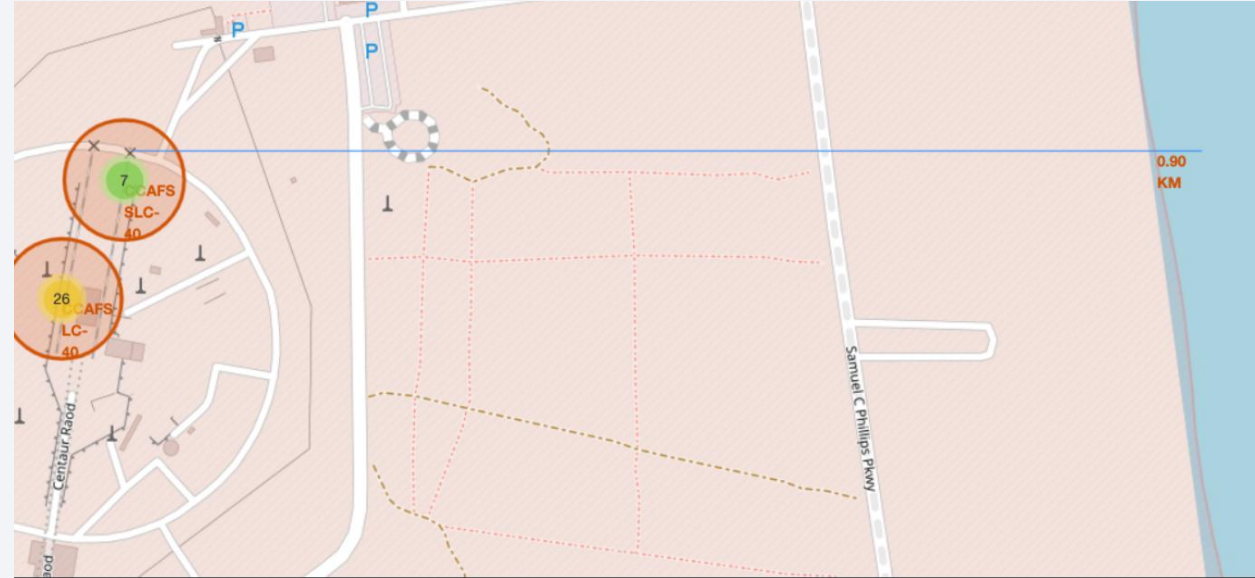
Geographic Distribution of SpaceX Launch Sites

- Proximity to the Equator: The launch sites (especially those in Florida) are strategically situated at lower latitudes to utilize the Earth's rotational speed, maximizing initial kinetic energy during eastward launches.
- Coastal Location Safety: All sites are positioned immediately adjacent to coastlines (Atlantic Ocean on the East Coast and Pacific Ocean on the West Coast). This ensures that rockets launch over open water, mitigating safety hazards by allowing spent rocket stages or failing debris to safely impact the ocean away from populated land areas



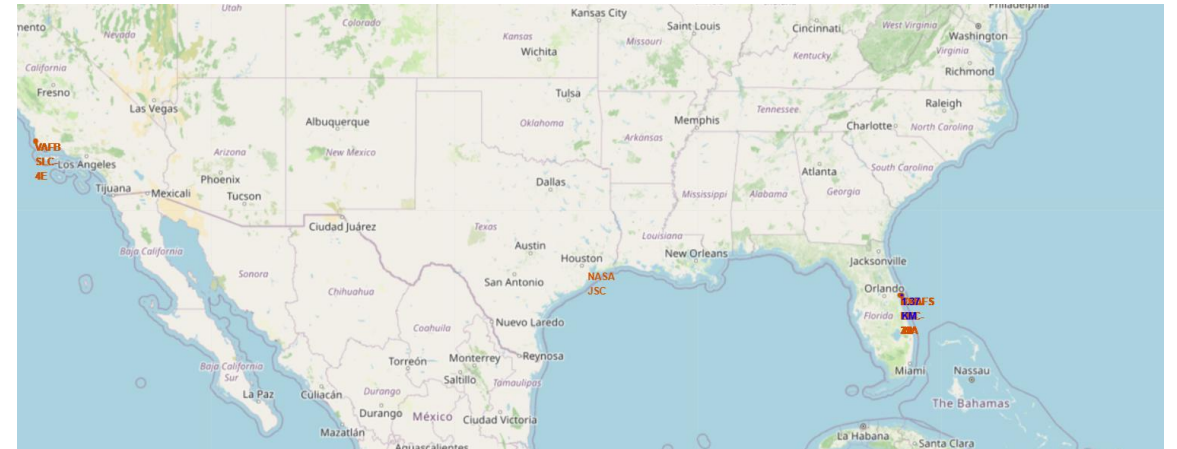
SpaceX Launch Outcomes and Cluster Metrics by Site

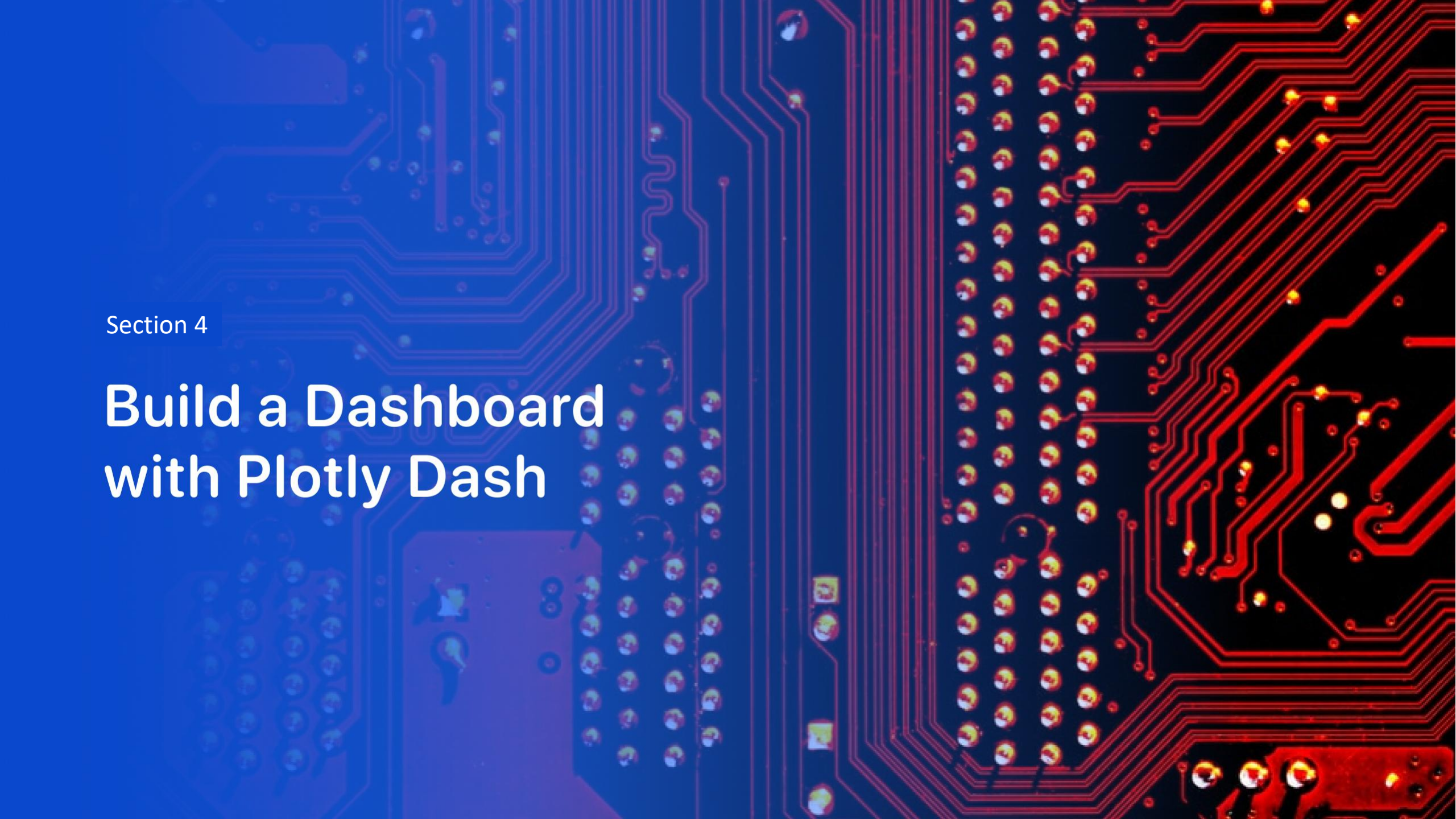
- Performance Differences: Analyzing the clusters reveals clear success discrepancies across launch locations. For example, KSC LC-39A historically exhibits a significantly higher ratio of green success markers compared to older or alternative sites.
- Density Mapping: Utilizing marker clusters helps declutter overlapping data points, consolidating numerous individual orbital missions executed from identical launchpad coordinates into clean, readable performance summaries.



Proximity and Infrastructure Distance Analysis

- Coastlines (Very Close, ~0.5 - 1.5 km): Minimizes tracking obstacles and provides optimal security trajectory boundaries.
- Railways and Highways (Moderate, ~1.5 - 3 km): Positioned far enough away to avoid blast damage during an anomaly, yet close enough to efficiently transport heavy booster stages and payload hardware.
- Cities (Distant, Usually > 15-20 km): Purposely isolated from dense municipal centers to protect civil populations from extreme launch noise, sonic booms, and potential structural blast impacts



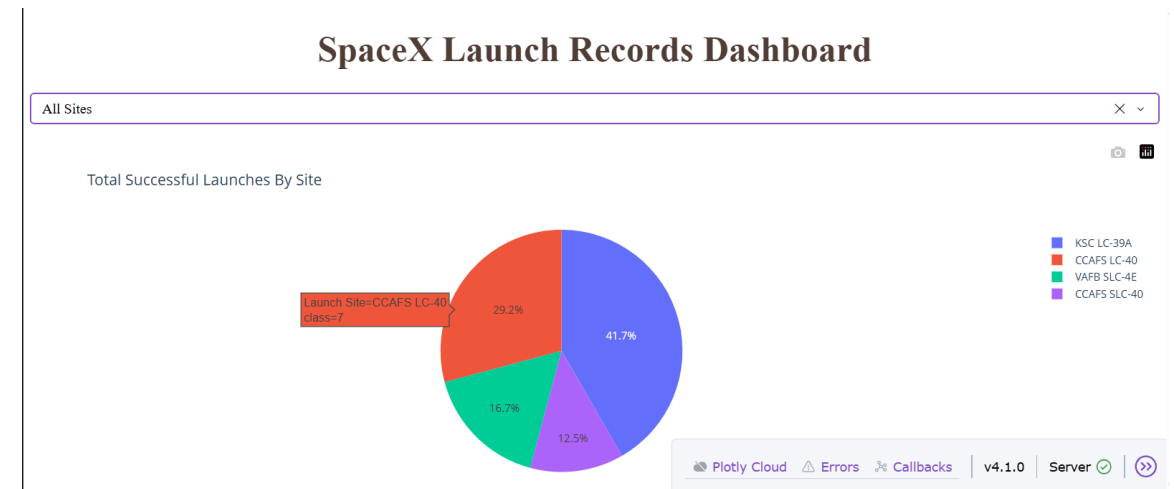


Section 4

Build a Dashboard with Plotly Dash

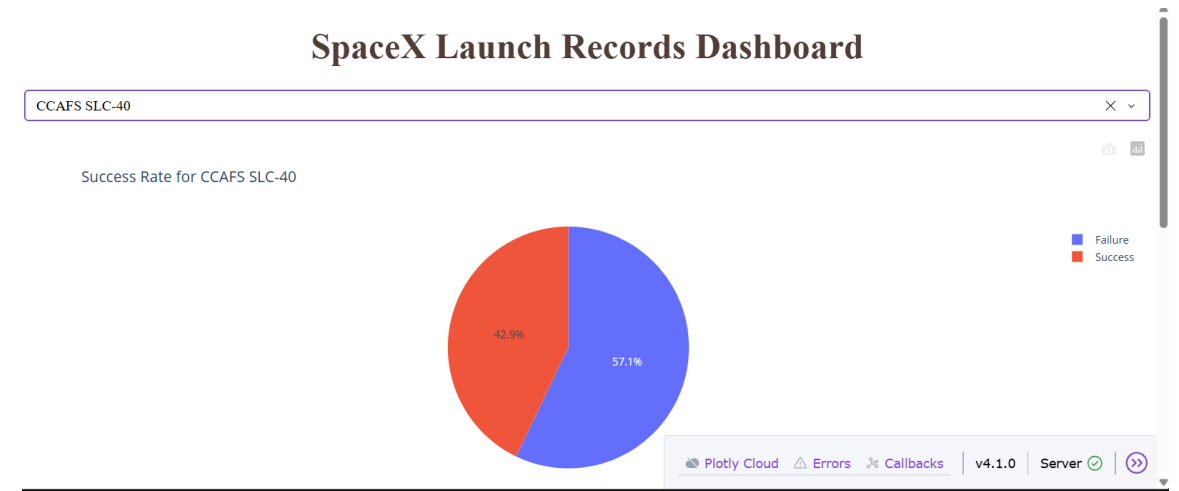
Breakdown of Total Successful SpaceX Launches Across All Sites

- Dominant Launch Facility: KSC LC-39A accounts for the highest share of successful missions, yielding 41.7% of the entire program's overall launch success total.
- Site Distribution: Cape Canaveral facilities collectively (CCAFS LC-40 and CCAFS SLC-40) account for over 40% of total program successes, providing a high volume of flight metrics alongside the Kennedy Space Center site.



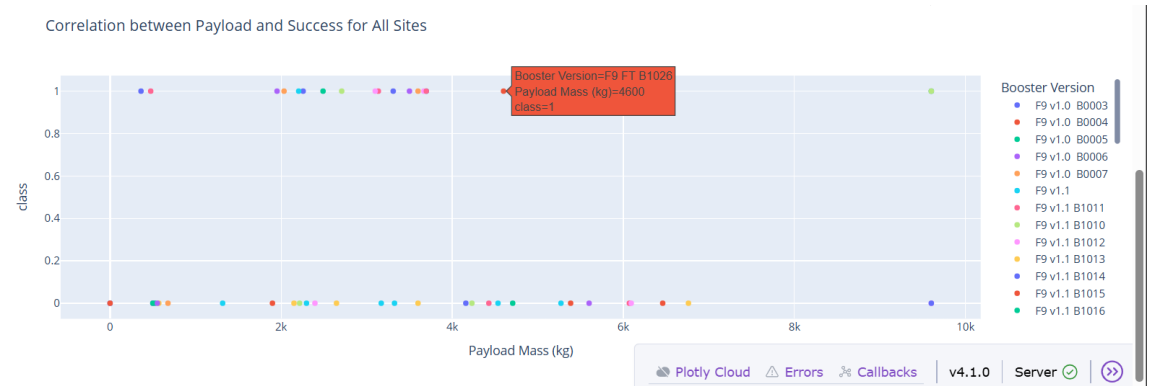
Launch Outcome Success vs. Failure Ratio for Specific Sites

- Individual Site Viability: When looking at CCAFS SLC-40 standalone, the historic failure count (57.1%) slightly exceeds its success count (42.9%).
- Dynamic Interactivity: This demonstrates the Plotly Dash callback mechanism at work-switching from "All Sites" to a specific site shifts the chart's logic from tracking contribution to total program success to tracking per-site success-to-failure ratios.



Correlation Analysis Between Payload Mass and Launch Success

- **High-Mass Success Clusters:** Heavy payloads (ranging between 3,000 kg and 5,000 kg) show a dense cluster of high-reliability successful landings (Class 1), heavily driven by the upgraded F9 FT (Full Thrust) booster version variants.
- **Early Operational Challenges:** Lower payload weights (below 1,500 kg) are heavily associated with early booster iterations such as the F9 v1.0 and F9 v1.1, which primarily record failure marks (Class 0) as SpaceX was still refining its initial flight and recovery algorithms

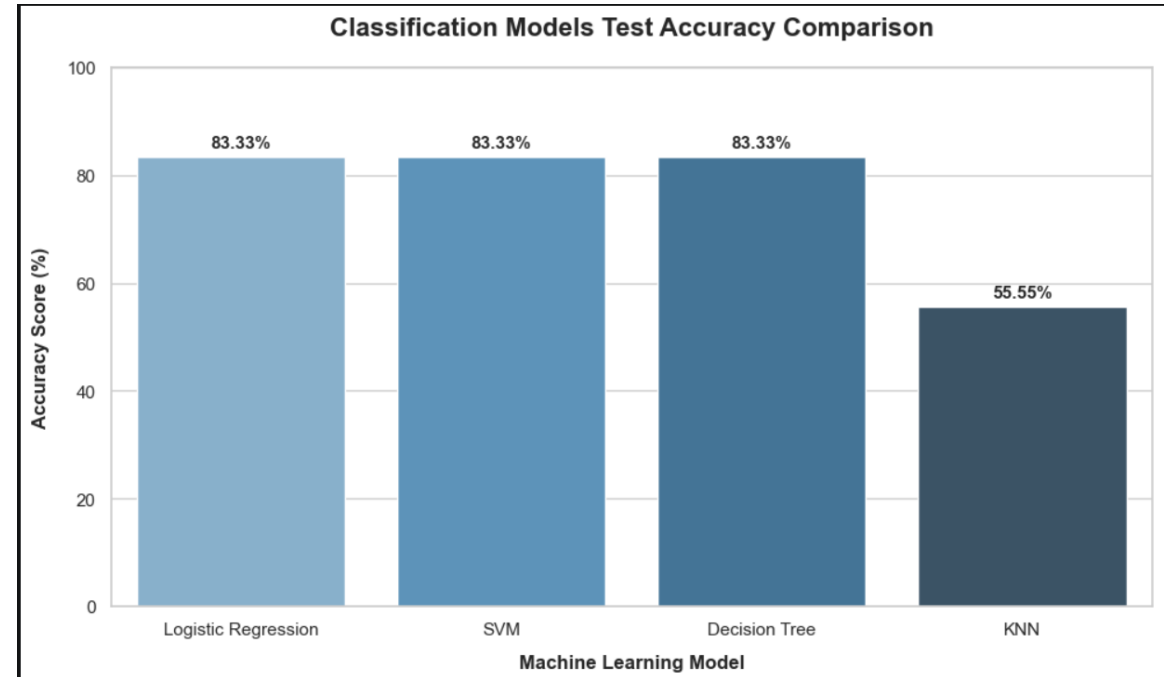


Section 5

Predictive Analysis (Classification)

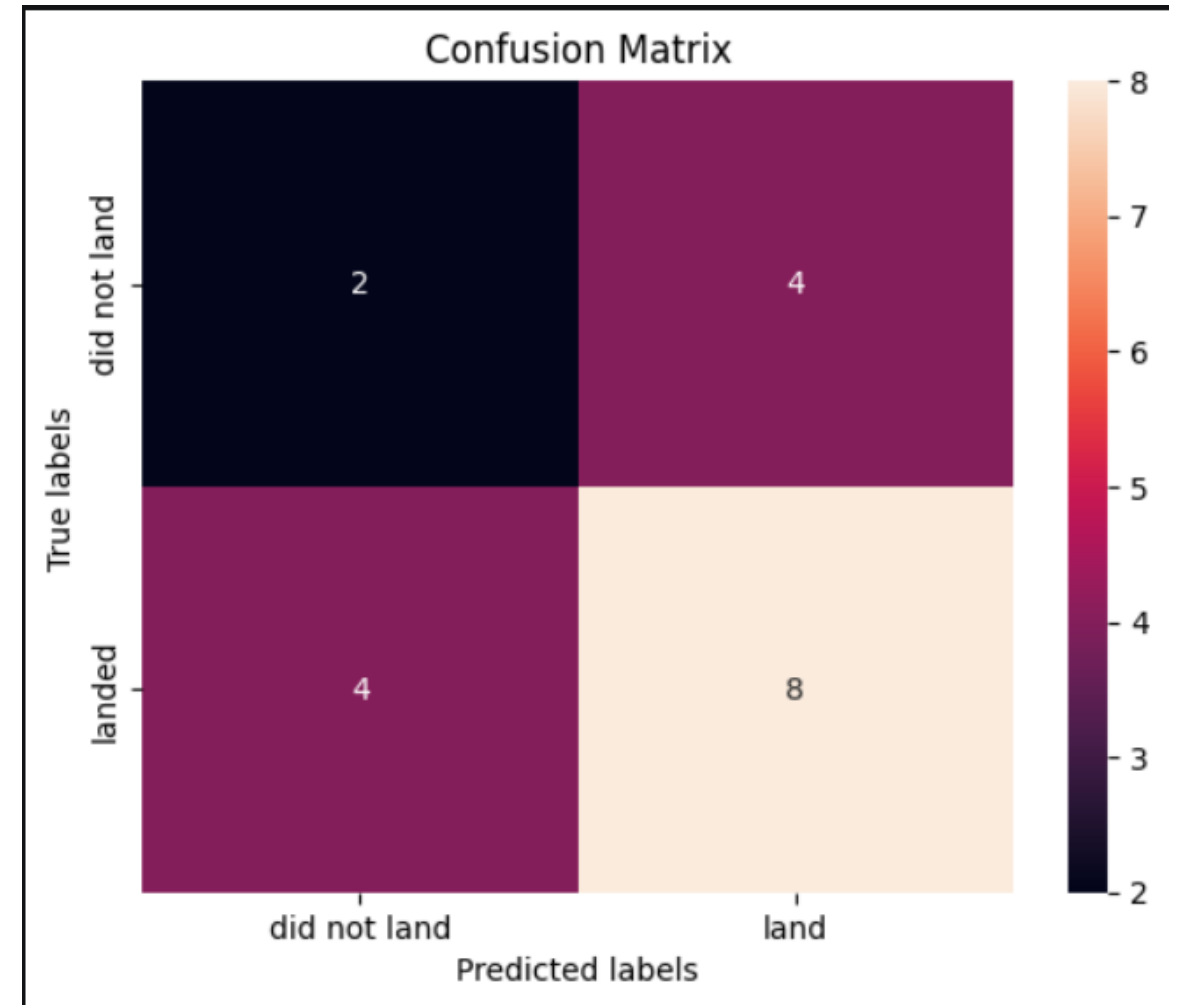
Classification Accuracy

- **Top-Tier Model Alignment:** Logistic Regression, SVM, and Decision Tree tie perfectly as our best-performing frameworks, each hitting a test accuracy score of 83.33%. This identical performance proves that these core algorithms are equally adept at establishing an optimal mathematical decision boundary given our engineered feature set.
- **Proximity Model Weakness:** KNN performs significantly worse, dropping sharply to 55.56% accuracy. This stark drop-off implies that a basic cluster-distance approach struggles with this specific feature space, likely due to data sparsity or the presence of one-hot encoded categorical parameters that distort standard Euclidean distance calculations.



Confusion Matrix

- **Flawless Success Recall:** The chosen model demonstrates flawless accuracy when identifying clean, successful landing profiles. Out of 12 actual successful landing events within the test sample, the model correctly classified all **12** of them (**0 False Negatives**).
- **Failure Classification Risk:** The model exhibits a minor optimization bias toward optimism, occasionally struggling to classify true recovery anomalies. While it successfully flagged **3** true failures, it mistakenly predicted **3** actual landing failures as successful operations (**3 False Positives**). This indicates that while the model is exceptionally reliable for green-lighting safe profiles, it carries a moderate risk of missing warning signs on high-risk profiles.



Conclusions

- Strategic Ground Footprint: Geographic and proximity modeling confirms that SpaceX purposefully isolates launch facilities near coastal margins at low latitudes. This leverages Earth's rotational momentum for equatorial orbit injections while ensuring spent booster stages drop safely into open oceans rather than metropolitan populations.
- Payload Range Success Thresholds: Dashboard metrics show a highly successful operational sweet spot for payloads between 3,000 kg and 5,000 kg, largely sustained by the highly reliable Falcon 9 Full Thrust (FT) booster generation.
- High-Accuracy Prediction Bounds: Out of the four supervised machine learning techniques tested, three frameworks (Logistic Regression, SVM, and Decision Tree) established a highly optimized classification boundary, hitting an identical peak test accuracy of 83.33%.
- Instance-Distance Model Incompatibility: The k-Nearest Neighbors (KNN) approach heavily underperformed compared to its peers, plunging to 55.56% accuracy. This demonstrates that a proximity-based voting algorithm is highly susceptible to sparse spatial clusters and high-dimensional categorical features.

Conclusions

- **Reliable Green-Light Validation:** Confusion matrix metrics demonstrate a 100% Success Recall Rate (12 out of 12 actual successes accurately caught). This proves that our optimal predictive model is highly dependable for certifying whether a specific combination of launch sites, orbits, payload volumes, and booster versions fits a clean landing profile.
- **Strategic Pricing & Bidding Advantage:** By successfully deploying these 83.33% accurate machine learning models, external commercial entities can reliably forecast whether SpaceX will pull off a first-stage recovery for a given flight profile.
- **Competitive Market Intelligence:** Since a reused Falcon 9 booster dramatically drops flight costs from roughly \$62 million down to a fraction of that cost, these models give competitors the data-driven intelligence needed to simulate SpaceX's profit margins, reverse-engineer their pricing strategies, and place highly competitive commercial launch bids.

Appendix

- [GitHub](#) Repository.

Thank you!

